

Teo Kai Jie (Kendrick)

📞 +65 8128 0249 | ✉ kj.teo.work@gmail.com | 🌐 kj-teovercelapp.vercel.app | 🌐 [kai-jie-teo](https://kai-jie-teo.vercel.app) | 🌀 [pixelhypercube](https://pixelhypercube.com)

EDUCATION

Nanyang Technological University

Bachelor of Computing (Honours) in Computer Science

Expected Jul 2028

Singapore

- **Current GPA: 4.56/5.00** (Highest Distinction)

Singapore Polytechnic

Diploma in Information Technology

Mar 2022

Singapore

- **Final GPA: 3.80/4.00**

EXPERIENCE

Singapore Armed Forces

Software Engineer (NSF)

Dec 2022 – Aug 2024, May 2025 – Jul 2025

Singapore

- Developed a **Node.js** and **EJS** email-based QR code attendance system to automate check-in for **300+ participants** in real time, earning a feature on Army News.
- Executed **QA testing** for Python-based Telegram chat-bots to improve system reliability by **~80%**.
- Designed new functional components for a Microsoft PowerApps manpower tasking application and built a dedicated mobile variant, resolving **300+ operational cases** after user testing.

EcoExchange Pte. Ltd.

Lead Frontend Developer (Industry Client Project)

Oct 2021 – May 2022

Singapore

- **Led a 3-man frontend team** to build a corporate intranet using **HTML5, CSS3, and JavaScript**.
- Digitized legacy paper-based operations into an interactive tracking portal, successfully processing **1,000+ live operational tickets**.

Leaptron Engineering Pte. Ltd.

Software Engineer (Intern)

Feb 2021 – Aug 2021

Singapore

- Deployed an **Android (Java)** QR code scanning tool for warehouse personnel, replacing manual data entry.
- Architected a **Node.js/MySQL** warehouse management system from scratch with normalized relational schemas.
- Created and tested **MySQL** stored procedures to handle inventory transactions, optimizing database execution times and improving tracking accuracy and efficiency by **~25%**.

PROJECTS

Personal Website | Website | Source Code

Next.js | TypeScript | Three.js | Vercel

- Developed an **LLM chatbot** using **Gemini API** and **ElevenLabs API** to stream real-time voice responses synthesized from a custom voice model.
- Built a **WebGL** 3D voxel viewer in **React Three Fiber**, using **InstancedMesh** batching and Intersection Observers to pause off-screen rendering and maintain **~60 FPS**.

CellCollab | Website | Source Code

React.js | Node.js | Socket.IO | Heroku

- Built a real-time multiplayer cellular automata playground using **Socket.IO** to enable **<100 ms** synchronization for live cell edits and active brush hover positions.
- Designed a **Node.js** simulation engine that executes state calculations at **100 ms/tick** and broadcasts board updates across **SHA-256** hashed rooms.

SKILLS

- **Languages:** TypeScript, JavaScript (ES6+), HTML5, CSS3, Java, Python, C++, C, C#
- **Frameworks:** Node.js, Express.js, React.js, Next.js, Android SDK, Tailwind CSS
- **Databases:** PostgreSQL, MySQL, MariaDB, MongoDB
- **Tools & Environments:** Git, GitHub